

Flash-Quard: Utilizing Advanced Digital Knitting Techniques to Develop Fashionable Reflective Garments

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Abstract

Merging protective, functional design with fashionable silhouette and surface design aesthetics, this design presents the development of a digitally knitted, reflective coat garment. This is executed by incorporating reflective materials into the textile design process using industrial grade digital knitting technology. Consideration for the shape and look of the garment took place alongside development of the functional efficacy of the reflective properties creating a cohesive, integrated design process. The garment uses a specialized knitted structure, a tubular jacquard for surface design execution along with added spacer yarn for textile weight and drape appropriate to a coat garment. Two sections of programming allow for a traditional contrasting-color plaid design to be combined with a reflective floral design that is embedded in the structure, visible under direct and concentrated light. This applied research follows research through practice framework, where both an identified need and inquiry into the capabilities of reflective knit structures combined in an iterative design process [1,2].

CCS Concepts

• Human-centered computing • Human computer interaction (HCI)

Keywords

Digital knitting; functional textiles; reflective apparel; user experience

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1 Introduction

The capabilities of industrial grade digital knitting technology have contributed to numerous innovations in a variety of industries, including architecture, medical, automotive, accessories, footwear, and apparel fields [3,4]. In apparel, this included products with embedded technology components [5,6]. These advancements have allowed for shaped, 3-dimensional textiles to be created in the shape of garments, allowing for new textile creativity, minimizing construction costs, and opening new avenues for dynamic garment pattern production. Here, we explore these capabilities by developing Flash-Quard, a high-visibility personal protective garment that utilizes protective, functional design without sacrificing aesthetics.

The form factor chosen for Flash-Quard was a coat, which was inspired by one author's experience living in a colder climate that experiences greatly decreased daylight hours during winter. The increased darkness necessitates habitual use of reflective apparel and accessories in daily life. Reflective apparel is often needed when walking as a pedestrian around town, and when walking to one's car through dark or dimly lit roads and parking lots. Pedestrian-car accidents are identified as a primary cause for pedestrian deaths [7,8], and visibility aids contribute to reducing this danger [9]. Traditional reflective garments prioritize function over style, which interrupts the average wearer's everyday ensemble. Therefore, this design offers a practical and stylish alternative without removing the function.

The coat blends the formal and the playful by incorporating reflective materials into the textile design process. This was achieved by following a research through practice framework [1,2], where an identified need (safety in low visibility environments) and production capabilities (digital knitting structures and processing) guide each other through an iterative design process. This allowed the textile structure, shape, and surface design to be developed at once, merging them into a cohesive garment. The final design uses a tubular knit jacquard with a spacer, filler yarn to add weight and warmth creates the passive body of the textile, while a traditional contrasting-color plaid design combined with a reflective floral design incorporate reflective yarns. By using the textile and yarns as a medium for executing the high-visibility function, we value care, comfort, and style that effortlessly incorporate into a user's current wardrobe while increasing their pedestrian safety.

2 Background

2.1 Personal Protective Equipment Design and Reflective Garments

Apparel is used to cover the body for both aesthetic and protective purposes [10]. As an aesthetic item it expresses and communicates something about the wearer to the others around them [11]. As a protective object, the level of protection can vary from privacy, environmental elements, and even chemical or biomedical [12]. Reflective apparel is a key form of personal protective equipment (PPE) that keeps the wearer visible in low-light or dark conditions by using bright colors or highly reflective materials to make the wearer stand out. Common places to find reflective PPE include uniforms for construction workers and jobs requiring outdoor work to meet OSHA industry safety standards [13], as well as garments for recreational activities such as running or cycling [14]. (Figure 1) In other applications reflectivity has been used to develop anti-paparazzi apparel that will conceal the wearer's identity by maximizing brightness and "blinding" a paparazzi camera [15]. High visibility products also include accessories such as wristbands and blinking lights attached via clips that can be added to clothing. (Figure 1) While these different formfactors achieve their functional purpose, they don't fully consider the need for visibility-enhancing PPE for the everyday pedestrian consumer. The construction or sporting goods aesthetic is not necessarily conducive to everyday wear, nor is the ultra-reflective blinding anti-paparazzi clothing. To encourage use of this type of PPE, it is important to create an aesthetic that appeals to the customer, as it is important to entice them to actually purchase and use the item [16].



Figure 1: Reflective product examples

2.2 Reflective Yarn

The development of specialized materials for textiles has grown significantly. Moisture-wicking, anti-bacterial, and even skin-rejuvenating yarns can be incorporated into textiles for a variety of apparel types. Here, we are naturally interested in retro-reflective yarns, as application of these yarns enable the creation of high-visibility garments. These can be found in a variety of colors, and require direct, concentrated light to be visible allowing them to blend into a background material when under normal lighted environments [12]. These capabilities are ideal for knit textile applications, as they can be engineered into the textile structure yet remain comfortable to wear. Another consideration for this material applied to a garment is the care; a coat is an outer

layer that is washed less frequently, which matches the care of the yarns. Prior studies have shown this type of metallic-coated yarn does require specific care to be used over prolonged periods. High heat and harsh detergents can dislodge the metallic coating and thereby dull the yarns over time [17] making this less practical for uniforms that would require intense and regular cleaning [12].

Creative design of reflective clothing using reflective yarn has been seen in previous research. One example is a project designed for the Ghanaian market which explored the application of reflective yarns to traditional African woven textiles. This project focused on surface design for visibility, executed in motifs that resonate directly with the local consumer, which has been established as critical for a user to want to purchase and use the item [9]. However, this work focused on woven rather than knitted textiles, and while testing confirmed the textile would be appropriate for garments commonly worn in the locale, the study did not expand to execution of actual garments. Another study focused on a creative exploration of reflective yarn application to create volume, relief, and shimmer. Here, the authors confirmed that reflective yarns, when applied in the correct structure and pattern, were effectively brilliant in bright light and offered expressive aesthetic design choices [18]. However, this project only advanced to the sample stage, and suggested apparel applications would be successful. In this project, we move beyond only exploring structures to apply them to a functional PPE prototype garment.

2.3 Digital Knitting

Digital knitting is a textile development process that can produce exact shapes in both flat and 3D forms, minimizing or eliminating waste and providing seamless support. Different stitch actions, loop lengths, number of needles activated, and partial knitting, or "goring", are used to create these shapes [19]. Combined with specialized yarns, a wide variety of products can be developed with specialized textile characteristics, introducing extensive possibilities for exploration. This can include wires, sensors, and other active components but these limit washability/care of garments and add a layer of difficulty for user interaction. Therefore, utilizing technical knitting we can incorporate PPE function and value into a textile at the structural level without adding additional weight, wires, or hardware that would change the normal wear of a garment. This allows us to address important considerations for functional garments like form language, placement, and texture in a user-friendly way [20,21], ensuring the design of the functional garment addresses both design and function at the same time during product development [25]. The reflective properties are embedded into the makeup of the garment through intentional design of reflective yarn placement, comfort of wearer, and shape of garment. Reflecting the personality of the wearer through design, while reflecting light to keep them safe from harm.

3 Design Development

The design process incorporated protective needs of the user, the capabilities of the digital knitting process, and the creative

design of the authors; in this way the objective is to create a garment with both symbolic and functional value and character [22]. The aesthetic properties of the reflective yarn were incorporated through surface design. Two different designs were used – a plaid and a floral repeating motif – to add impact to the reflective properties. (Figure 2) The plaid was chosen to be featured on the coat as it would be an expected repeating pattern for this type of garment. For the reflective yarns, a bolder, more graphic surface design was created to be knitted into the background using reflective and non-reflective yarns of the same color. This made the floral motif a secondary surface design that would only be visible when needed for reflective purposes, adding an element of playful surprise to the high visibility function.

3.1 Yarn Selection

The yarn selection first considered the function and protective priorities of the garment. To effectively incorporate the reflective properties of the retro-reflective yarn, a supporting yarn with a weight and hand similar to the reflective yarn was needed to both balance and blend the two yarns invisibly. Previous research shows successful use of natural fibers with retro-reflective yarns [23]; therefore, cotton and silk fibers were selected for testing. First, a raw, slubbed silk in a color matching the reflective yarn was plated with the reflective yarn, creating the “invisible” daylight appearance of the floral pattern, while cotton added body in the grey and could also be matched to the reflective yarn in the areas the motif crosses the plaid. These options supported the purpose of the coat’s reflective qualities, but additional weight was needed for a coat garment. Therefore, an air-spun spandex was added as a spacer, or “filler” material that would be knitted in between the front and back beds of the knitting machine. This added the weight and thickness needed to the textile structure without affecting the function of the surface design.

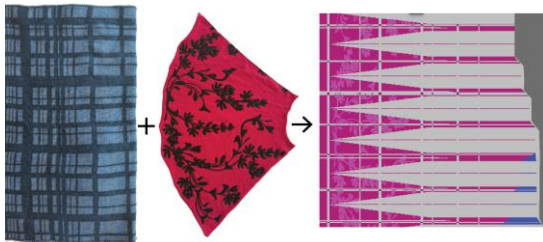


Figure 2: Repeating Plaid and Floral Surface Design Motifs Blended in the Programming, with 3-D shaping

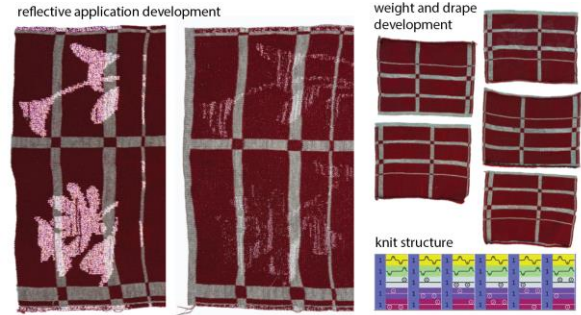


Figure 3: Samples to develop structure and surface design

3.2 Structure and Surface Design

This exploration began with the goal to incorporate grey plaid stripes with a red background with the red reflective motif knitted “invisibly” into the background of the plaid. The structure was developed through tests and iterative editing. It began with a basic structure that would allow three different colors/yarns to be featured in the blended surface design: the grey plaid stripe, the brick red background, and the reflective yarn. The iterative process is done by repeatedly moving between file programming and knitting to develop gauge swatches. These swatches inform exploration of yarns, knit structure, stitch length, and how each influences the possible design outcome. These samples progressed in the following order:

1. Twill-back jacquard with 3-colors in each section. This sample was too open, and the reflective yarn becomes heathered and minimally effective. The weight was not appropriate for the coat garment.
2. Tubular jacquard with 3-colors in each section. This sample was too open, and the reflective yarn becomes heathered and minimally effective. The weight was not appropriate for the coat garment.
3. Tubular jacquard with 3-colors in each section, with added full needle spacer. The surface design was divided into brick red and grey sections based on the plaid design. The reflective yarns were plated (set over or under) the non-reflective yarns but created holes in the garment based on the spacer structure. However, the reflective properties were successful.
4. Tubular jacquard with 3-colors in each section, with added 2-courses of 1x1 spacer. The surface design was divided into brick red and grey sections based on the plaid design. The reflective yarns were plated (set over or under) the non-reflective yarns. The spacer edit provided the correct weight, and the reflective properties were successful. This became the final structure design.

The successful gauge sample was then used to program the shapes for the final garment. One surprise that did occur with this sample is the reflective quality of the wrong side of the textile. In response, we consider the construction of the piece to allow it to be reversible. You can see both back and front reflective properties in Figure 5.

3.3 Shape and Silhouette

The coat silhouette is a double-breasted, raglan sleeve coat with a flared hem; subtle 3-dimensional shaping at the bust flows into a cinched waist, leading into a full flare below the waist. Because this shaping requires goring, the surface design is interrupted by the short-row technique of shaping. Therefore, the plaid stripes were carefully matched across the goring sections, and Adobe Illustrator was used to map out the reflective floral design to create uninterrupted repeating reflective motifs across the goring. This map was imported into the programming software and blended with the plaid stripe. (Figure 2)

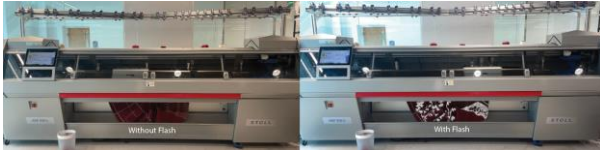


Figure 4: Knitting the Garment



Figure 5: Fully-fashioned sleeve with integrated cuff

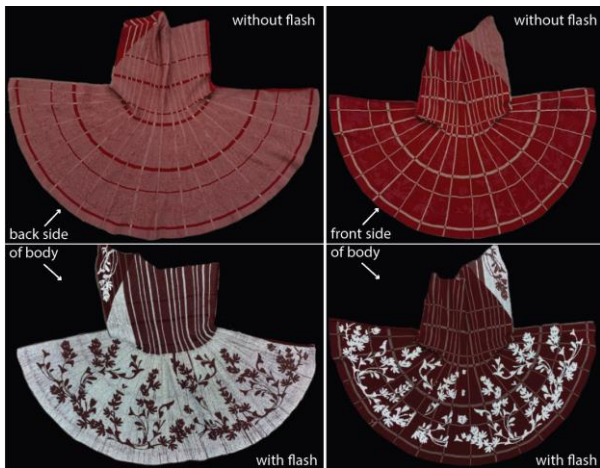


Figure 6: Body shape knitted, with reflective properties tested

3.4 Execution

Using the knowledge gathered from the sampling process, the coat was programmed as fully-fashioned, or shaped, sleeve and body pieces and knitted on a Stoll ADF 830 KI W Knit & Wear 7.2 gauge. (Figure 4) The body pieces are knitted sideways, with the knitted shape starting at the overlapping outside edge of the

placket. Goring is used to create a 3-dimensional shape for the bust and skirt flare, and the collar knitted in with the surface design reversed to show on front when folded. (Figure 6) To ensure the textile is balanced, these pieces are knitted in two mirrored shapes, so the loop structure is mirrored. The raglan sleeves feature an integrated cuff with the surface design treatment reversed to show when folded up like the collar. (Figure 5) A view of the final garment with the reflective component hidden by daylight versus revealed in a photo taken with flash can be seen in Figure 7.



Figure 7: Final Flash-Quard garment in multiple light conditions

4 Conclusion

By merging needs of the everyday user, the capabilities of the digital knitting process, and an iterative design process, the resulting coat keeps the user safe and visible as they walk and explore in dark environments. It is aesthetically attractive and provides warmth in cold temperatures, making it accessible for easy integration into an existing wardrobe to encourage use. In this process both knit garment design knowledge along with technical programming knowledge underpin this work. Application of digital knit technology is often separated between designer and programmer/technician [24]; here, by eliminating these boundaries the simultaneous knowledge of design, materials, structure, and programming enabled application of embedded functional and protective materials into the knitted garment simultaneously.

Future work will focus on different reflective yarn color and motif placement to test if color and placement could enhance visibility and aesthetics. Also, while the focus of this design project is to work towards creating a high visibility garment for the everyday pedestrian as proof of concept, we hope to compare the effectiveness of the reflective coat with reflective standards for outdoor construction worker safety in a future work to ensure our design achieves its goals.

References

- [1] Elizabeth Bye. 2010. A Direction for Clothing and Textile Design Research. *Clothing and Textiles Research Journal* 28, 3 (2010), 205-217. DOI:https://doi.org/10.1177/0887302x10371505
- [2] Chanjuan Chen and Kendra Lapolla. 2019. The Exploration of Geometric Modular System in Textile and Apparel Design. *Clothing and Textiles Research Journal* 39, 1 (2019), 39-54. DOI:https://doi.org/10.31274/itaa.8771
- [3] Mirela Blaga, Daniel Dorin, Ramona Ciobanu, and Dorin Ionesi. 2011. Interactive application for computer aided design of 3d knitted fabrics. In *Proceedings of eLearning & Software for Education Conference, Bucharest, Romania*, 1-8. Accessed June 20, 2025 <https://proxy.library.kent.edu/login?url=https://search.ebscohost.com/login.aspx?direct=true&AuthType=ip&db=e5h&AN=64924766&site=eds-live&scope=site>.
- [4] Wonseok Choi and Nancy B. Powell. 2005. Three dimensional seamless garment knitting on V-bed flat knitting machines. *Journal of Textile and Apparel, Technology and Management* 4, 3 (2005), 1-33.
- [5] Hongci Hu, Mengqi Jiang, Kai Lin, Kinor Shou-xiang Jiang, and Ziqian Bai. 2025. ReKnit-Care: A Seamless-Knitted Sensing Glove for Sensory Rehabilitation and Adaptive Haptic Feedback. In *Proceedings of the Nineteenth International Conference on Tangible, Embedded, and Embodied Interaction (TEI '25)*. Association for Computing Machinery, New York, NY, USA, Article 58, 1–7. <https://doi.org/10.1145/3689050.3705979>
- [6] Irmandy Wicaksono, Don Derek Haddad, and Joseph Paradiso. 2022. Tapis Magique: Machine-knitted Electronic Textile Carpet for Interactive Choreomusical Performance and Immersive Environments. In *Proceedings of the 14th Conference on Creativity and Cognition (C&C '22)*. Association for Computing Machinery, New York, NY, USA, 262–274. <https://doi.org/10.1145/3527927.3531451>
- [7] Ali S Al-Ghamdi. 2002. Pedestrian-vehicle crashes and analytical techniques for stratified contingency tables. *Accident Analysis & Prevention*. 34, (2002) 205-214. DOI:https://doi.org/10.1016/s0001-4575(01)00015-x
- [8] Wen Hu, Jessica B. Cicchino, An examination of the increases in pedestrian motor-vehicle crash fatalities during 2009–2016, *Journal of Safety Research*. 67, (2018), 37-44. DOI:https://doi.org/10.1016/j.jsr.2018.09.009.
- [9] Raphael Kanyire Seidu, Sun Young Choi, and Shouxiang Jiang. 2023. Development and performance of jacquard woven retro-reflective textiles with African design patterns. *Fashion and Textiles*. 10, (2023) 1-20. DOI:https://doi.org/10.1186/s40691-022-00322-8
- [10] Lou Taylor. 2004. *Establishing Dress History*. Manchester University Press.
- [11] Yuniya Kawamura. 2018. *Fashion-ology*. Bloomsbury Publishing.
- [12] Susan M. Watkins and Lucy Dunne. 2015. *Functional Clothing Design*. Bloomsbury Publishing USA.
- [13] Breakdown of OSHA. *Waste Advantage*. Retrieved June 20, 2025 from <https://wasteadvantagemag.com/breakdown-of-oshas-high-visibility-regulations-for-construction-workers/>
- [14] Ellen Ming. 2023. The Rise of Reflective Apparel: Market Insights and Industry Trends. *LinkedIn*. Retrieved June 20, 2025 from <https://www.linkedin.com/pulse/rise-reflective-apparel-market-insights-industry-trends-ellen-meng--1xvnc>
- [15] Liz Stinson. 2015. Paparazzi-Proof Clothing That. *Wired*. Retrieved June 20, 2025 from <https://www.wired.com/2015/02/paparazzi-proof-clothing-thats-embedded-reflective-glass/>
- [16] Seifert, Christin, and Veena Chattaraman. "Too new or too complex? Why consumers' aesthetic sensitivity matters in apparel design evaluation." *Journal of Fashion Marketing and Management*. 21, 2 (2017) 262-276. www.emeraldinsight.com/1361-2026.htm
- [17] Lee, Suhyun, and Sohyun Park. "Optimizing washing conditions for smart fabrics: a comprehensive study." *RSC advances*. 14, 54 (2024): 40098-40116. DOI:https://doi.org/10.1039/d4ra07365g
- [18] Nesrin Önlü and Havva Halaçeli. 2012. Visual Effects of Stainless Steel and Retro-Reflective Yarns on Woven Fabrics. *Research Journal of Textile and Apparel* 16, 2 (May 2012), 82–95. DOI:https://doi.org/10.1108/rjta-16-02-2012-b009
- [19] Su Liu, Siyao Mao, and Peihua Zhang. 2021. Toward on integrally-formed knitted fabrics used for automotive seat cover. *Journal of Engineered Fibers and Fabrics*. 16, 1-12, (2021). DOI:https://doi.org/10.1177/155892502111006544
- [20] Dunne, L. E., & Smyth, B. (2007, April). Psychophysical elements of wearability. In *Proceedings of the SIGCHI conference on Human factors in computing systems* (pp. 299-302).
- [21] Francine Gemperle, Christopher D. Kasabach, John M. Stivorc, Malcolm Bauer, and Richard Martin. 1998. Design for wearability. In *International Symposium on Wearable Computers*, October 19, 1998. IEEE Computer Society, 116–122. <https://doi.org/10.1109/ISWC.1998.729537>
- [22] Elsa Rosenblad-Wallin. 1985. User-oriented product development applied to functional clothing design. *Applied Ergonomics*. 16, 4 (1985), 279–287. DOI:https://doi.org/10.1016/0003-6870(85)90092-4
- [23] Raphael Kanyire Seidu and Shouxiang Jiang. 2023. Analysis of performance properties of retro-reflective woven fabrics made with retro-reflective yarns and natural yarns. *Textile Research Journal*. 94, 5-6, (2023) 533-551. DOI:https://doi.org/10.1177/00405175231202786
- [24] Adriana Gorea, Fatma Baytar, and Eulanda A. Sanders. 2021. Challenges and design opportunities in prototyping seamless knitted apparel: a case study. *International Journal of Fashion Design, Technology and Education* 14, 2 (2021), 127–138. DOI:https://doi.org/10.1080/17543266.2021.1881170
- [25] Pettys-Baker, R., Clarke, M. E., & Holschuh, B. (2024, October). Functional Now, Wearable Later: Examining the Design Practices of Wearable Technologists. In *Proceedings of the 2024 ACM International Symposium on Wearable Computers* (pp. 71-8)